EART11300 Practical and Professional Skills

Week 6 – Folds and Faults on Geological Maps

Introduction

Most of your work to date on geological maps has considered relatively flat-lying deposits or those where the rock layers (strata) are arranged conformably, in sequences from oldest at the base to youngest at the top. Today, our focus is on folded and faulted rocks – which are the result of tectonic forces or stresses acting on the solid crust. The overall aims of the class are to be able to:

- Describe the geometry of folds and faults within geological strata;
- To understand the tectonic processes, or geological properties of the rocks themselves, that cause folds and faults to develop;
- To be able to identify folds and faults on geological maps, and describe their geometry in simple terms.

Describing the Orientation of Rock Strata using Dip and Strike

How do we describe the orientation of geological strata? Note that "strata" are just layers of rock, the word for an individual layer is "stratum". One way to describe strata orientation is using **strike** and **dip**.

The **strike** of some planar surface, such as a bedding plane, is *the compass bearing of the imaginary line formed by the intersection of the planar surface with a horizontal surface*. A strike line is always horizontal. The compass bearing is measured from north (000°) and reported in three digits (e.g. 145°).

The **dip** of a planar surface, e.g. the same bed, is always measured at 90° to strike and is *the angle measured down from horizontal*, so will always a number between 0° (horizontal) and 90° (vertical). To avoid ambiguity, dip direction should be indicated with a compass point, e.g. 145 / 50 SW.

Dip and dip direction is another method of showing the orientation of planar surfaces on maps. Dip is the same as in the strike and dip method. Dip direction is the compass bearing of the direction of dip projected to the horizontal. For example: a geological bed dips at an angle of 30 degrees towards the south. This would be written as 30/180 (because the compass bearing for south is 180°).

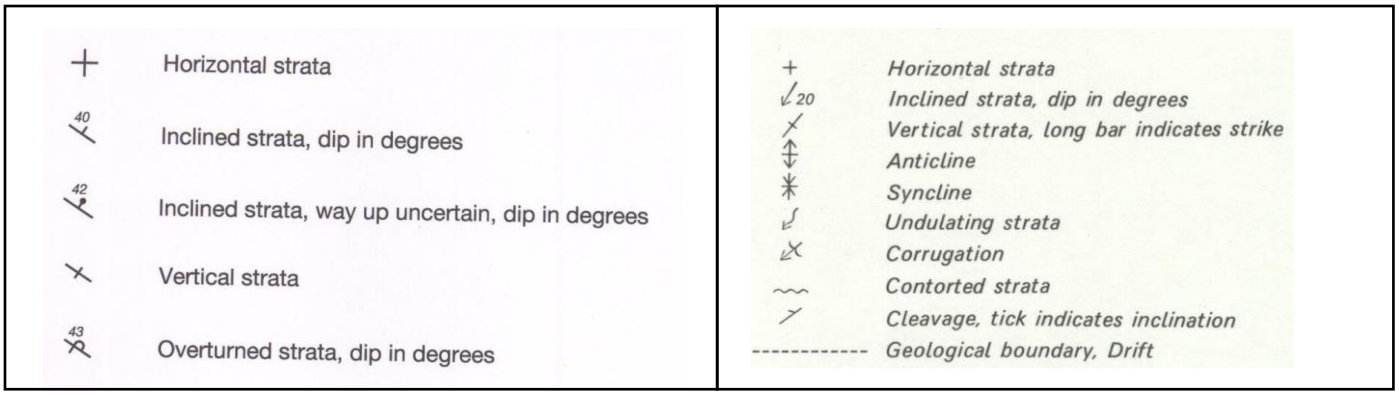
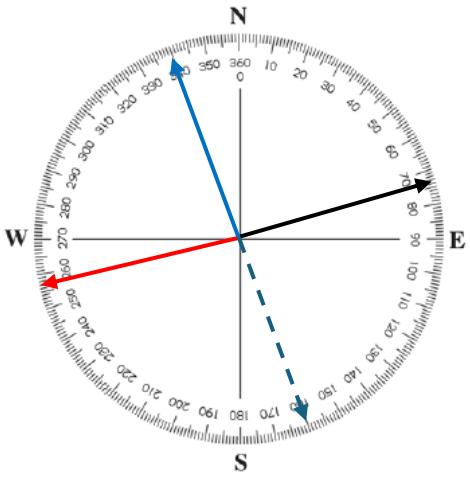


Figure 1. The left panel shows modern British Geological Survey map icons, where strike is the longer line and dip is the short line with a numerical label. Right panel shows older (though still common) notation, where the arrow represents dip direction. The number in both examples represents the dip angle.

Exercise 1. Strike, dip, and dip direction.



a) Draw the strike line on the compass dials for a bed striking 075 – what would another acceptable reading for this strike be?

Black arrow shows 075. Red arrow shows 255 (075 + 180)

Ideally using a second colour, draw the strike line for a bed striking 160. What would another acceptable reading be?

Dashed arrow shows 160. Blue arrow shows 340.

b) Now consider a bedding plane, that has a strike and dip of 085/40SE. Can you rewrite this as a dip and dip direction?

Dip and dip direction = **40/175** – In this case you add 90° to the strike direction to get the correct dip direction. Look at the compass and check that 175 is towards the SE

Now consider a bedding plane that has a strike and dip of 085/40NW. Can you rewrite as dip and dip direction?

Dip and dip direction = **40/355** – In this case you add 270° to the strike direction to get the correct dip direction. Look at the compass and check that 355 is towards the NW.

c) Figure 2 shows an area cropped from the British Geological Survey map of Pembrokeshire. Find some icons that show the strike and dip, or dip and dip direction of the strata. Can you describe the overall range in dip and dip direction of the bedding? How does strike vary across this area?

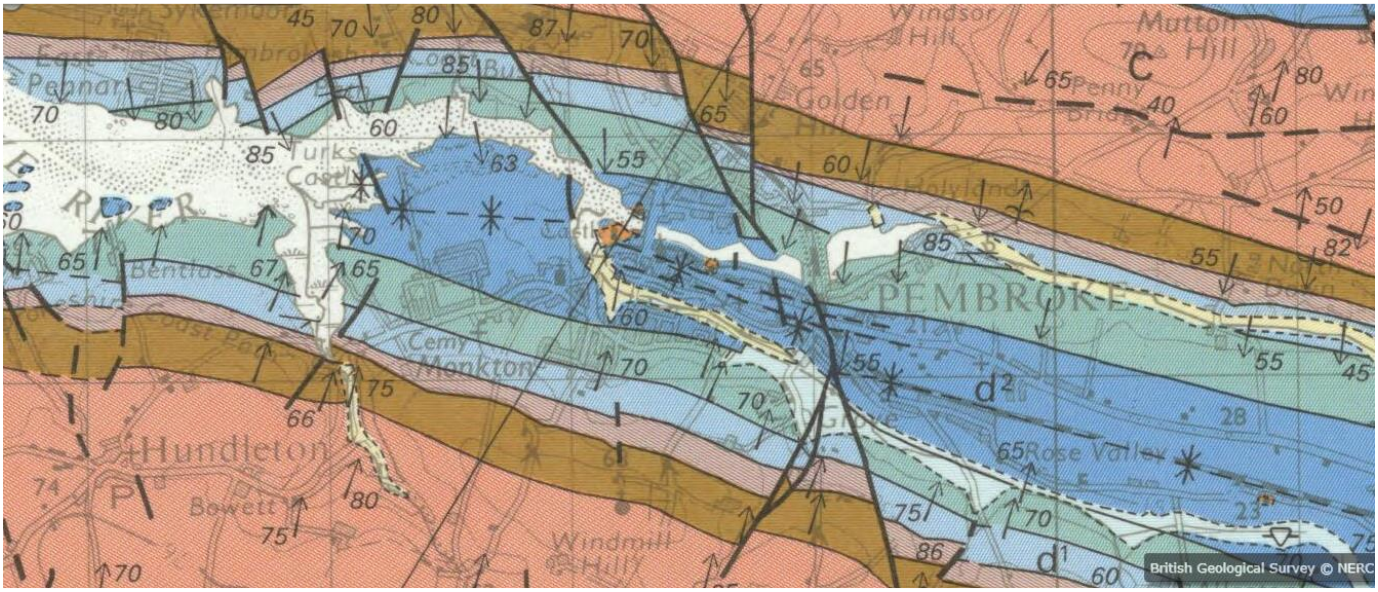


Figure 2. Cropped region from the BGS map of Pembrokeshire. Dip and dip direction are marked with labelled arrows. The arrows point in the downdip direction.

There is a wide range in dip value, from vertical and very steep (>70°) to more moderate, but still rather steep dips (30-55°). There are numerous changes in dip direction across the map, with most beds dipping broadly either north or south. Beds in the lower half of the image are dipping north (?inward); those in the upper half are dipping south. Most beds are striking roughly east-west. Bonus points if you infer / realise that this is a folded structure…

Folds and Folded Rocks

Folded rocks are the consequence of compressive forces within the Earth's crust, and represent a ductile mode of crustal shortening. Remember that ductile deformation is folding and brittle deformation is faulting (which we'll come to later).

First, let's learn some essential terminology. A folded surface which closes downwards is referred to as a **synform,** one which closes upwards is an **antiform**. A fold in which the youngest rocks are in the core (the inside part) is called a **syncline**; a fold with the oldest rocks in the core is an **anticline**. Synforms and antiforms are therefore purely geometric descriptive terms, while anticlines and synclines imply we have some geological knowledge about the relative ages of the beds.

The Y-shaped icon shown in Figure 3 is called a younging symbol. The tail of the 'Y' points towards the youngest strata. It is always drawn at 90° to the strike of the beds. Ask a demonstrator if you are unsure about this!

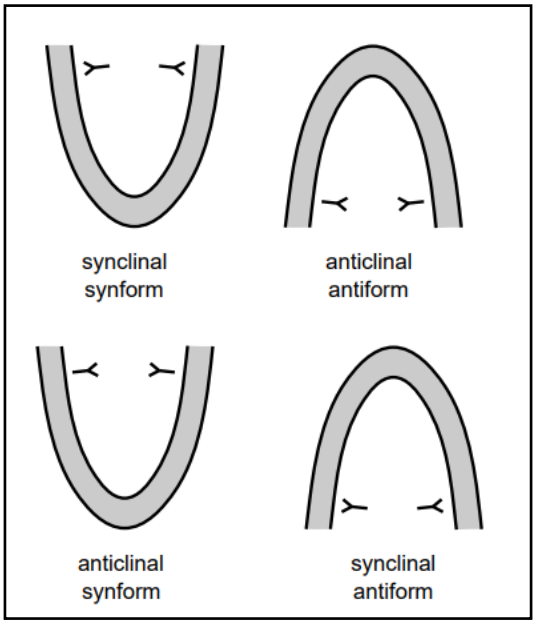


Figure 3. Some simple folds.

How can we describe folds more fully? Look at Figure 4:

- a fold has two **limbs** formed from the bedding planes that are dipping in different directions;
- the two limbs meet at a **fold hinge** where the bedding changes from one orientation to another;
- the imaginary line along the fold, parallel to the hinge, is called the fold axis;
- the imaginary plane that can be drawn through the hinge line (and fold axis) that bisects the profile of the fold is called the axial plane;
- the angle between the limbs is, very pleasingly, called the interlimb angle.

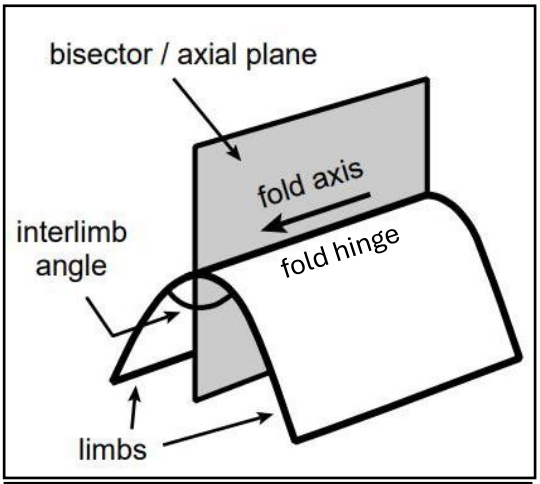


Figure 4. Antiform, illustrating key fold terminology.

We can then characterise different folds according to the variation in their geometry (Figure 5).

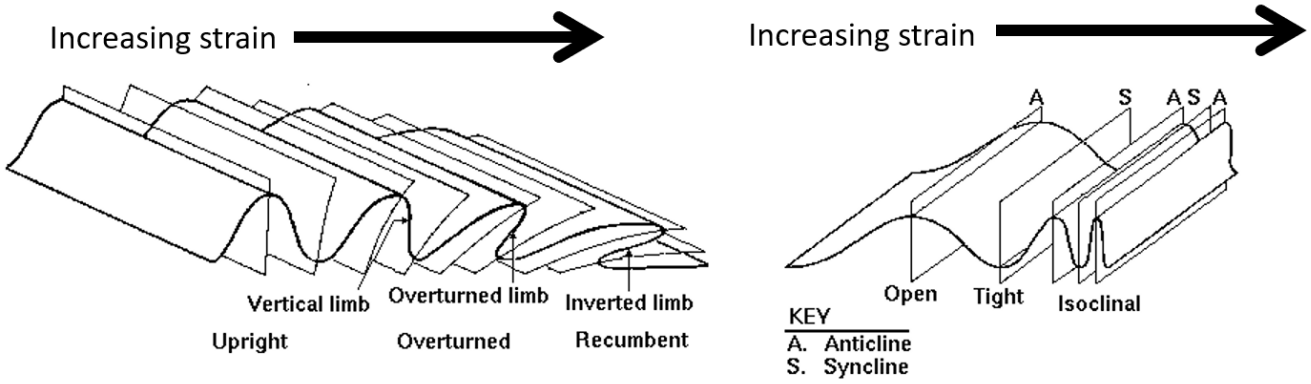


Figure 5. The orientation of the axial plane (left) and the variation in the interlimb angle (right) are important variables for describing fold geometry.

Exercise 2. Getting Familiar With Folds

Use the modelling clay provided on the benches to form models of flat-lying geological strata, with at least two layers and if you can, three. You should decide which colour represents the oldest strata, and which colour represents the youngest. In small groups, experiment with forming different kinds of fold by bending your initially flat strata into new shapes.

Firstly, make an anticlinal antiform.

Draw a sketch of your fold in the box to the right and label it to show:

- Limbs
- fold hinge
- younging direction
- axial plane.

Estimate the interlimb angle and the dip of the axial plane.

Now, modify your model so that this anticlinal antiform becomes both isoclinal and recumbent. Describe in writing, and illustrate with a second sketch, how the dip of the axial plane and interlimb angle have changed.

The axial plane in an upright anticlinal antiform may be 90° or least dipping rather steeply. If the fold is relatively open, the interlimb angle may be rather large, e.g. 45-50°.

The axial plane in a recumbent fold will have a shallow dip, close to horizontal, and if the fold is also isoclinal, the dip angle will be low, ~10-20°.

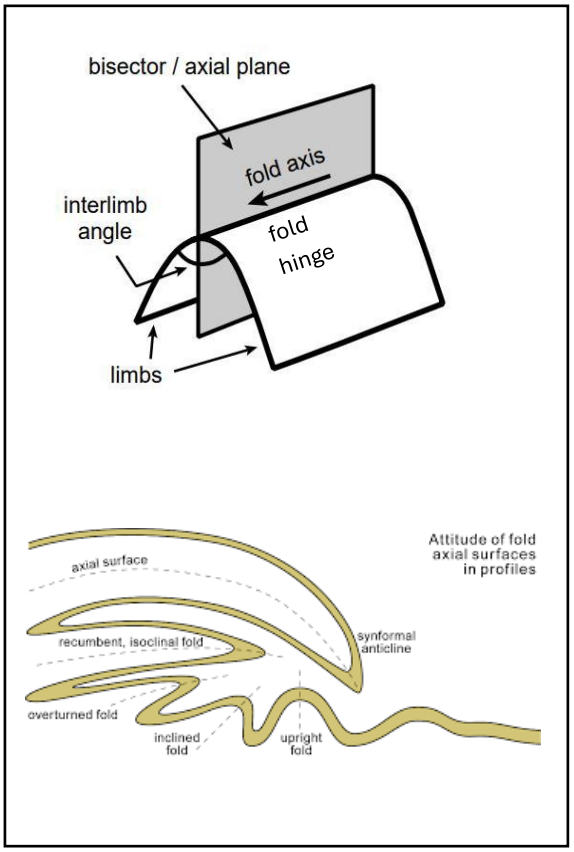


Figure 6. Sketches of an upright anticlinal antiform (above) and a recumbent isoclinal fold (below). Note change in angle of fold axial planes (represented by dashed lines)

Exercise 3. Visualising Folds on a Geological Map

The exercise above is helpful for understanding fold geometry and descriptive terminology, but how will folds appear on the map, when effectively viewed from above? To begin, remake your simple anticlinal antiform. When viewed from above, all you will be able to see is the youngest layer of rock and it is not easy to know that a fold is present. Luckily, out in the real world, erosion is a powerful force for exposing bedrock at different levels in the subsurface.

Using a university ID card (or other thin piece of plastic), slice the uppermost layer off your model fold – ask a demonstrator for help if you are unsure how to do this. Viewed from above, what does your fold now look like? It should appear similar to the sketch in Figure 7, showing parallel bands of rock, which are repeated outwards from the fold axis.

For anticlines, older rocks will appear in the centre of the structure, in map view, depending on the depth of erosion (Figure 7a). For synclines, the reverse is true: the youngest rocks appear in the centre of the structure and the rocks to either side are increasingly older (Figure 7b).

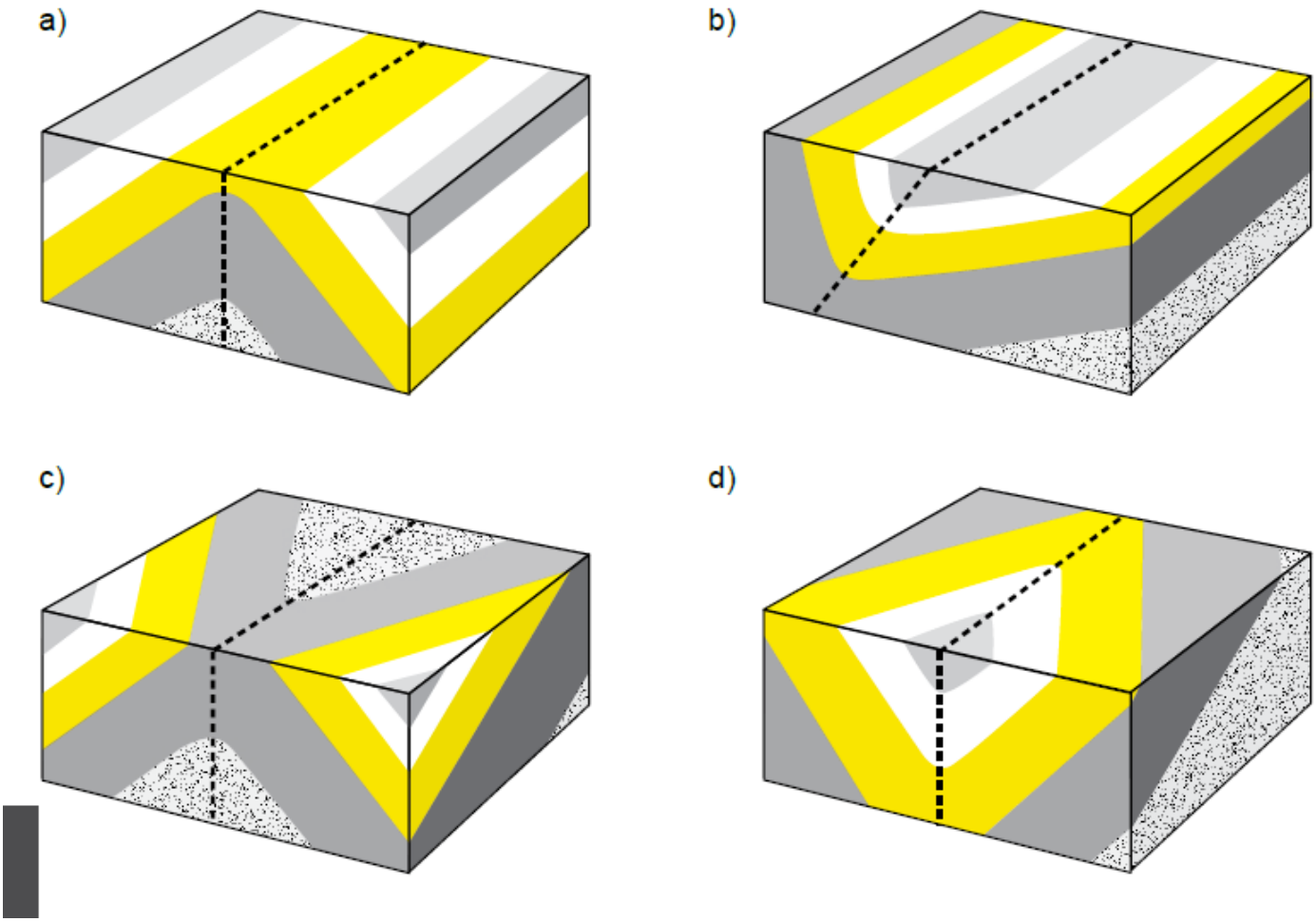


Figure 7. Box models showing both cross-sectional views and map views of folded strata, where age order is from black speckled (oldest) to mid-grey to yellow to white to pale grey (youngest). (a) non-plunging anticline with relatively old rocks exposed in the centre; (b) non-plunging syncline with youngest rocks exposed in the centre; (c) plunging anticline; (d) plunging syncline. Courtesy of J. Houghton (U. Leeds).

Before moving on, carefully remake your simple anticlinal antiform for a final time. This time, tilt it slightly so that the fold axis is dipping rather than horizontal. We call the dip of a fold axis the **plunge**. Now, slice across the top of the fold as before, and note how the outcrop pattern (in map view) has changed as a result of the fold plunging, rather than having a horizonal fold axis. This is challenging to visualise and is something that some of your will study in greater detail in second year. It is important for all of you, however, to realise that folds can plunge and that this changes the outcrop pattern (see Figure 7c and 7d for further illustrations). One key different (compare Figure 7a and 7a) is that much older rocks can be exposed in the centre of the structure if the fold is plunging.

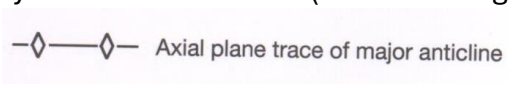
Now look back at Figure 2, taken from the geological map of Pembrokeshire. Hopefully it is clear to you now that this area contains a large folded structure. If the blue unit ('d2') is the youngest, and using the dip directions to guide you, determine whether this is a synform or antiform, and a syncline or anticline. Can you tell whether or not it is plunging?

Youngest in the core = syncline. Dipping inward = synform. This is a synclinal synform. Plunge is hard to assess. Beds are reasonably parallel, apart from furthest west. The fold may close to the west but it's ambiguous from this small view. Dips don't change (if plunging, they'd change along fold axis).

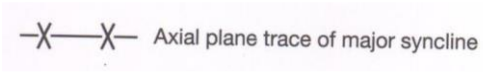
Exercise 4 – Axial Traces

To finish our work on folds, consider the sketch map below (Figure 8), showing a map view of some folded structures. Four different rock units are shown, from oldest (1) to youngest (4). Identify three folded structures on the sketch map and mark their axial trace. The axial trace is a line projected onto the flat surface of a map that runs parallel to the fold axis.

When marking an antiform, remember that the dip direction on the limbs is away from the fold axis. Therefore, we annotate the axial trace with diamond shaped symbols attached to it (think of triangles pointing away from the fold axis):



When marking a synform, the dip direction on the limbs is towards the fold axis. Therefore, we annotate the axial trace of a synform with bow-tie shaped symbols attached to it (think of triangles pointing towards the fold axis):



Can you determine whether each of these structures represents a syncline or an anticline?

From west to east, there is a syncline (youngest in core), anticline (oldest in core), syncline.

Given that we lack information on the dip direction, we cannot say for sure whether they are also synforms or antiforms. Which do you think is most likely? To figure this out, try to draw a sketch cross section along a west-east line in Figure 8. Start from the central bed of unit 1 – decide whether the beds of unit 2 on either side going to dip inwards, or outwards? There are two possible solutions here, both of which are feasible but one of which is more likely geologically. Ask for guidance if you don't know where to start. We will go through this exercise as a class before the end of the lesson today.

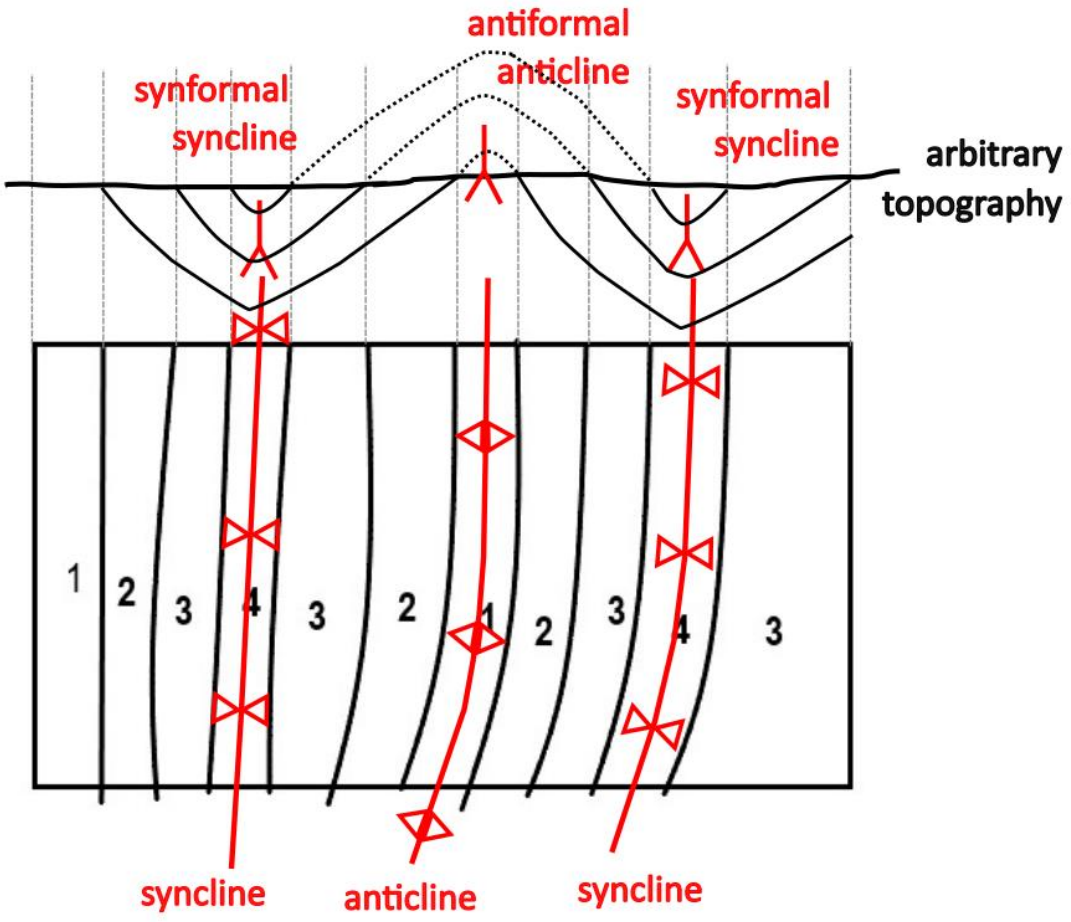
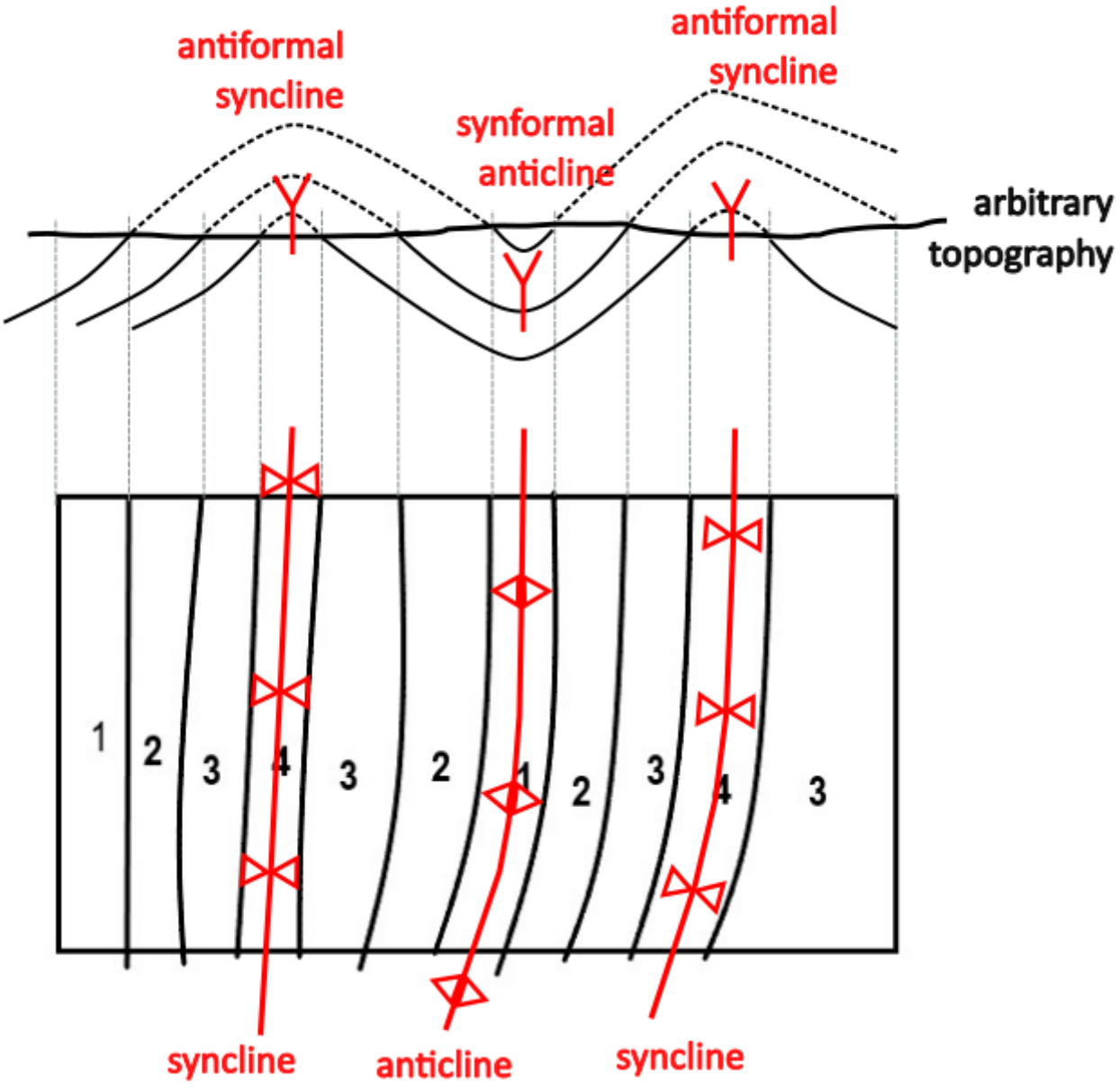


Figure 8. Sketch map of three folded structures.

Plausible, but less likely? Requires
full overturn of beds/structure



Faults and Faulted Rocks

Faults are formed, instead of folds, when rocks undergo brittle (instead of ductile) deformation. This can be due to the material properties of the rocks involved, or because the rocks in question are at lower depths and temperatures (higher pressure and temperature favours ductile deformation), or also because of high strain rates (rapid deformation) which favours brittle deformation.

A fault can be defined as a planar surface within the crust along which rocks have been displaced. Different kinds of faults are found in different tectonic settings and are characterised by different kinds of displacement. In the lecture, you have seen something like Figure 9, a classification scheme for faults according to the orientation and relative magnitude of principal stresses (forces) acting on the rocks.

To fully characterise a fault we need to know (1) the orientation of the fault plane surface, (2) the orientation of the slip vector (which describes the movement direction) and
(3) the amount of displacement.

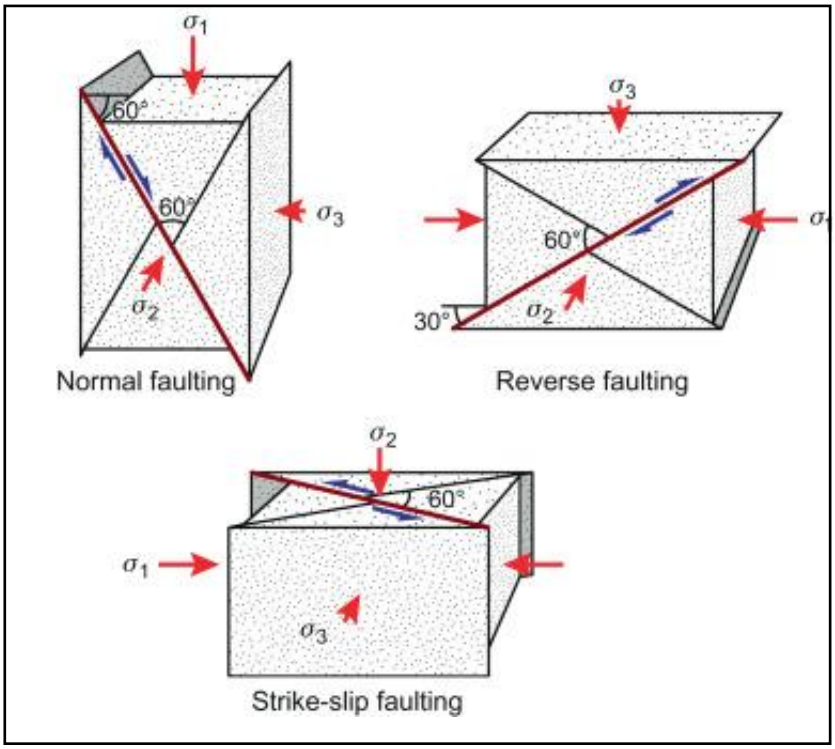


Figure 9. Illustrations of the main types of fault, defined by orientations of maximum (σ_1), minimum (σ_3) and intermediate (σ_2) principal stress (after Anderson, 1905).

Today we're mostly concerned with distinguishing three main types of fault, based on their geometry and the sense of motion:

- normal faults have steeply dipping fault planes (~60°) and the hanging wall block ("H" in Figure 10a) has moved down (is "downthrown") along the fault plane. This has the effect of placing young rocks (white colour in Figure 10a) against, or on top of, older rocks (greys in Figure 10a).
- reverse faults may also have steeply dipping fault planes, similar to normal faults, but they can also be much shallower (~30°), in which case they are called thrust faults. The hanging wall in a reverse or thrust fault moves up vertically and this results in older rocks (grey in Figure 10b) being placed on top of younger rocks (white in Figure 10b).
- both normal and reverse faults are called 'dip-slip' faults because they have a strong vertical component to their displacement
- strike-slip faults involve mostly lateral/horizontal displacements, though can involve movement to the left or the right (imagine standing on one side of the fault plane and looking across, to see in which direction the rocks have moved).

The presence of types of fault in a particular place can help to build up tectonic history of an area. Across the UK for example, large ~SW-NE thrust faults reveal how the country has been assembled from a number of ancient crustal fragments. Some of you will return to this topic in more detail later.

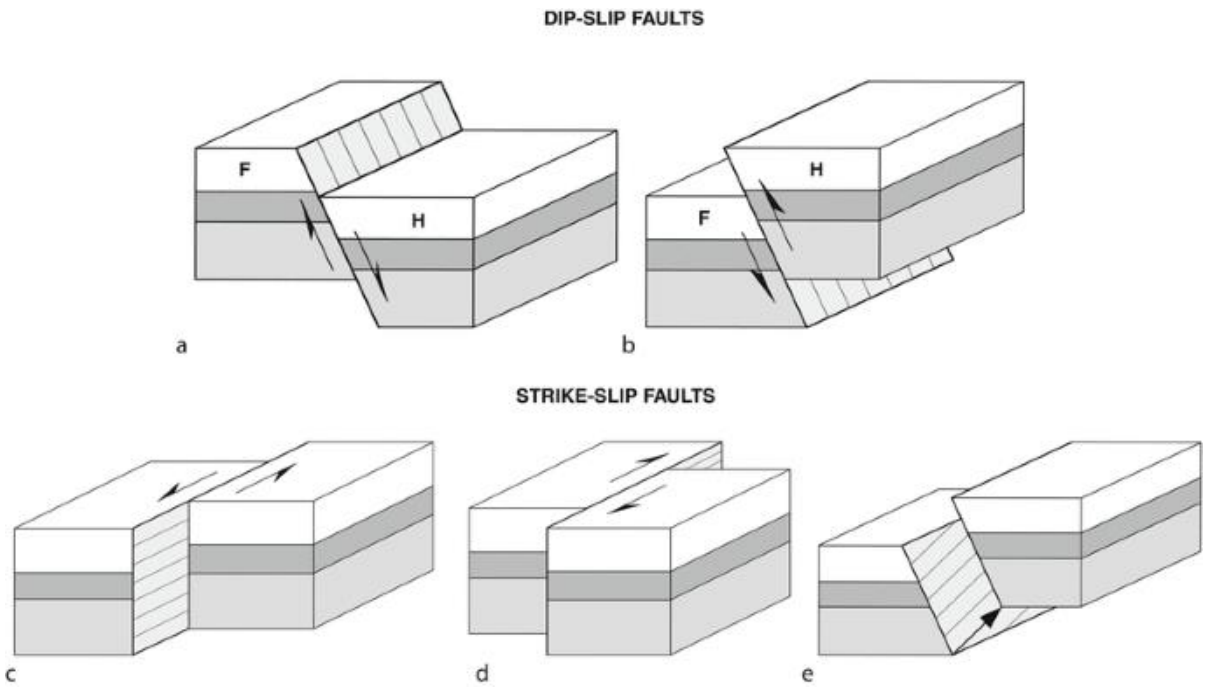


Figure 10. Sketches showing dip-slip (a=normal, b=reverse) and strike-slip faults (c=left lateral; d=right lateral; e=oblique slip.

Fault Type	σ_1 Orientation (Max Stress)	σ_2 Orientation (Intermediate Stress)	σ_3 Orientation (Min Stress)	Tectonic Setting
Normal Fault	Vertical (downward)	Horizontal	Horizontal (extension direction)	Rift zones, divergent boundaries
Reverse (Thrust) Fault	Horizontal (compression direction)	Horizontal	Vertical (upward)	Convergent boundaries, mountain belts
Strike-Slip Fault	Horizontal	Vertical	Horizontal (perpendicular to σ_1)	Transform faults, shear zones

Figure 11. Reference guide to different fault types, defined according to the principal stress orientations of Anderson (1905, 1951).

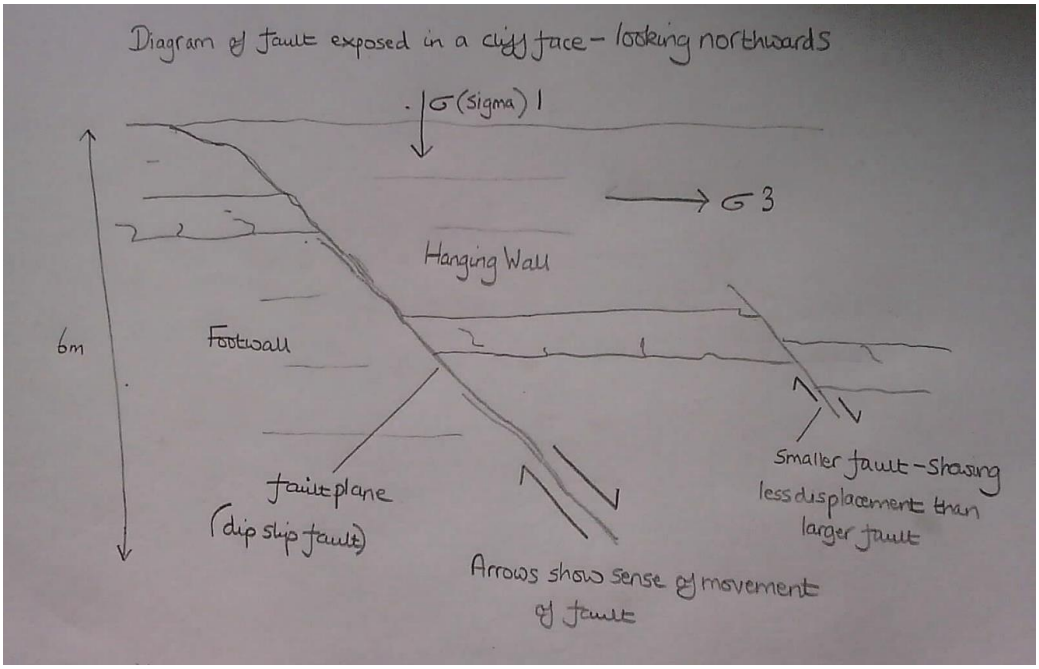
Exercise 5 – Field Sketches of Faulted Rocks

In the following pages, you are provided with two photographs of faults exposed in the field. The first is in sedimentary rocks of lower Cretaceous age, in the La Herradura Formation, Morro Solar, Peru. The second is also in sediments, but of Jurassic age strata of the Curtis and Entrada Formations, Utah.

In your sketches, try to capture the main structural elements that define the faults. Look first for geological **bedding**, especially 'competent' or hard rocky beds and use these to highlight the **displacement direction and magnitude** across the fault. You could note if there is an apparent variation in **rock type** and whether different rocks behave differently in the fault. You should also try to identify the location of the **fault plane** and note the **orientation and dip angle**. Your sketch should include **scale** (note the people shown in each image). Finally, annotate your sketch showing **footwall**, **hanging wall**, and the directions of the three **principal stresses**.



Figure 12. La Herradura Formation, Peru. Wiki Commons.



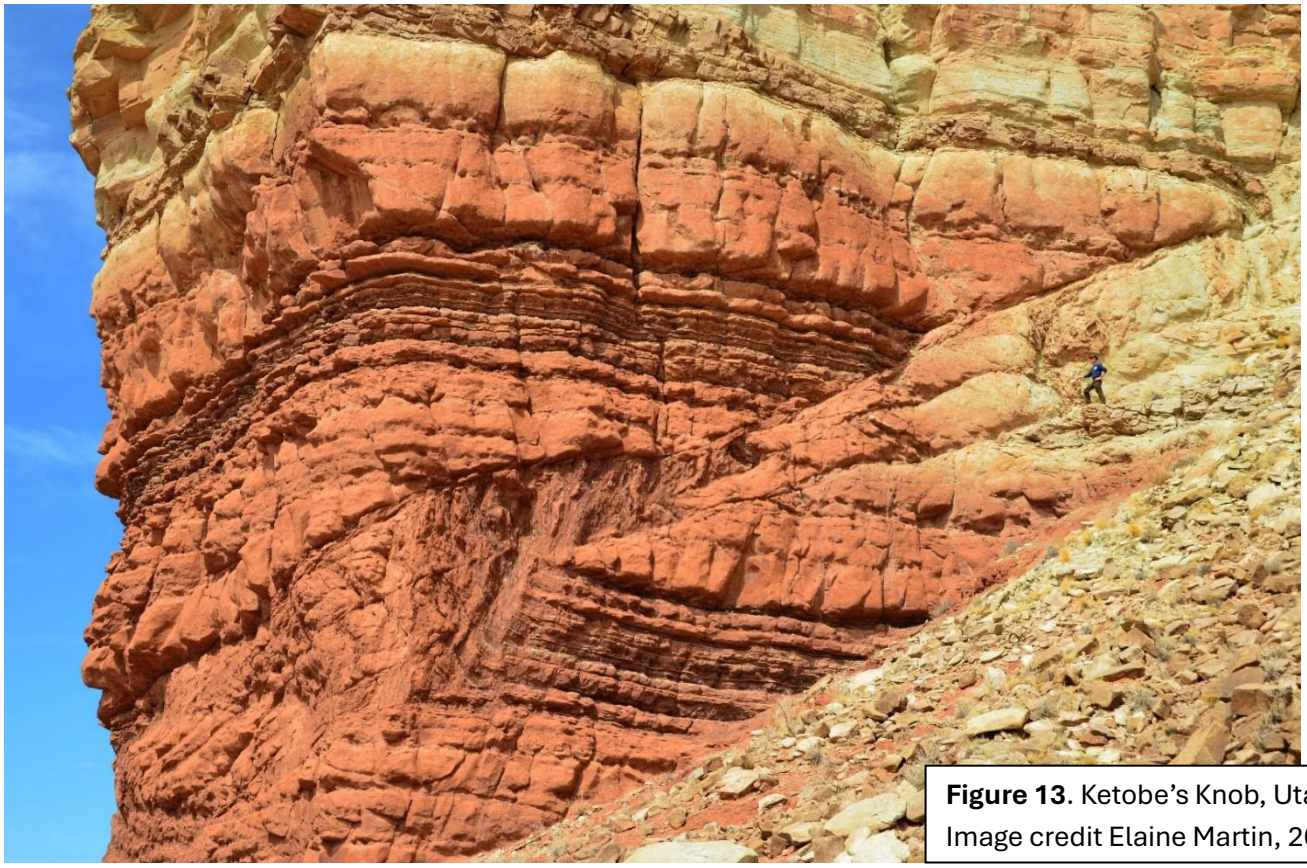
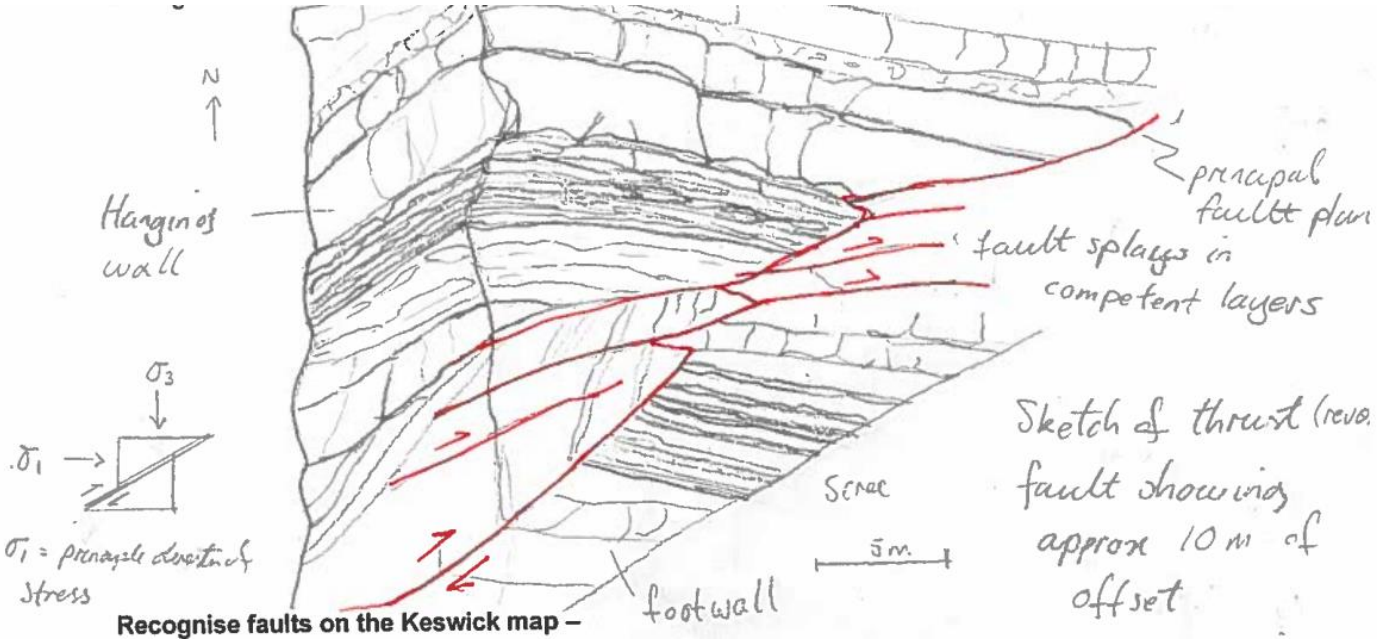


Figure 13. Ketobe's Knob, Utah. Image credit Elaine Martin, 2008.



The first image shows a normal fault, with steep dip; the second is a thrust with shallow dip. There is a lot more detail to be seen, if the students are motivated, especially in the second image but this is tricky and probably not adequately covered in lectures yet: could point out the fault splays, ramps, etc.

Exercise 6 – Faults on the Keswick Map

Now we will explore how faults appear on geological maps. At this stage, we will take advantage of the fact that different symbols are used to represent different types of fault. Figure 14 shows the different lines used to represent faults on British Geological Survey maps.

What is most important to note is that these symbols give an indication of displacement direction across or along faults, and help us identify which side of the fault is the hanging wall or footwall. If we also use the colour of rock types and their relatives ages, obtained from the strasigraphcic section on the maps, we can understand why faults are marked as normal, thrust, or strike-slip.

Study the BGS map of the Keswick area. You can find it online at the BGS map portal (https://www.bgs.ac.uk/information-hub/bgs-maps-portal/) but today you should look at the printed copies of the map provided on the benches.

Find the Causey Pike Fault, the Derwent Water Fault, and the Coniston Fault. The first two are close to the centre of the map, and the third is at the lower centre. Using the lines with which these are drawn, you should be able to identify whether these are normal, thrust or strike-slip faults. In each case, use geological evidence (e.g. ages of strata) from the map to justify these designations.

What type of fault is the Causey Pike Fault? Along the western parts of its outcrop the Causey Pike Fault is a thrust fault and in the eastern part of the map it has changed to a strike slip fault. The diamond symbols ("teeth") on the fault contact are on the hanging wall side of the fault. Use the teeth to figure out the hanging wall and the stratigraphic column to determine the age juxtaposition. You should be able to show older rocks have been placed on top of younger: Ordovician rocks of the Kirkstile Formation (KSt) are in the hanging wall, i.e. on top of the older Buttermere Formation (BUF) in the footwall. To the east, these rock formations are again adjacent to each other but it is hard to see reliably what the horizontal displacement may be.

What type of fault is the Derwent Water Fault? This is a strike slip fault – the arrows show the sense of movement. Encourage the students to try to determine whether it is left or right lateral. They could also try to determine how much displacement has occurred. It is not easy to see – look for the orangey "vs" unit towards the southern end of the fault as the best evidence.

What type of fault is the Coniston Fault? This is a normal fault. The small "tick" symbols are on the downthrown hanging wall side of the fault plane. Try to find examples of younger rocks displaced downwards in the hanging wall placing them in contact with older rocks in the footwall. This is also relatively complex as the downthrown side changes from north to south along the length of the fault!

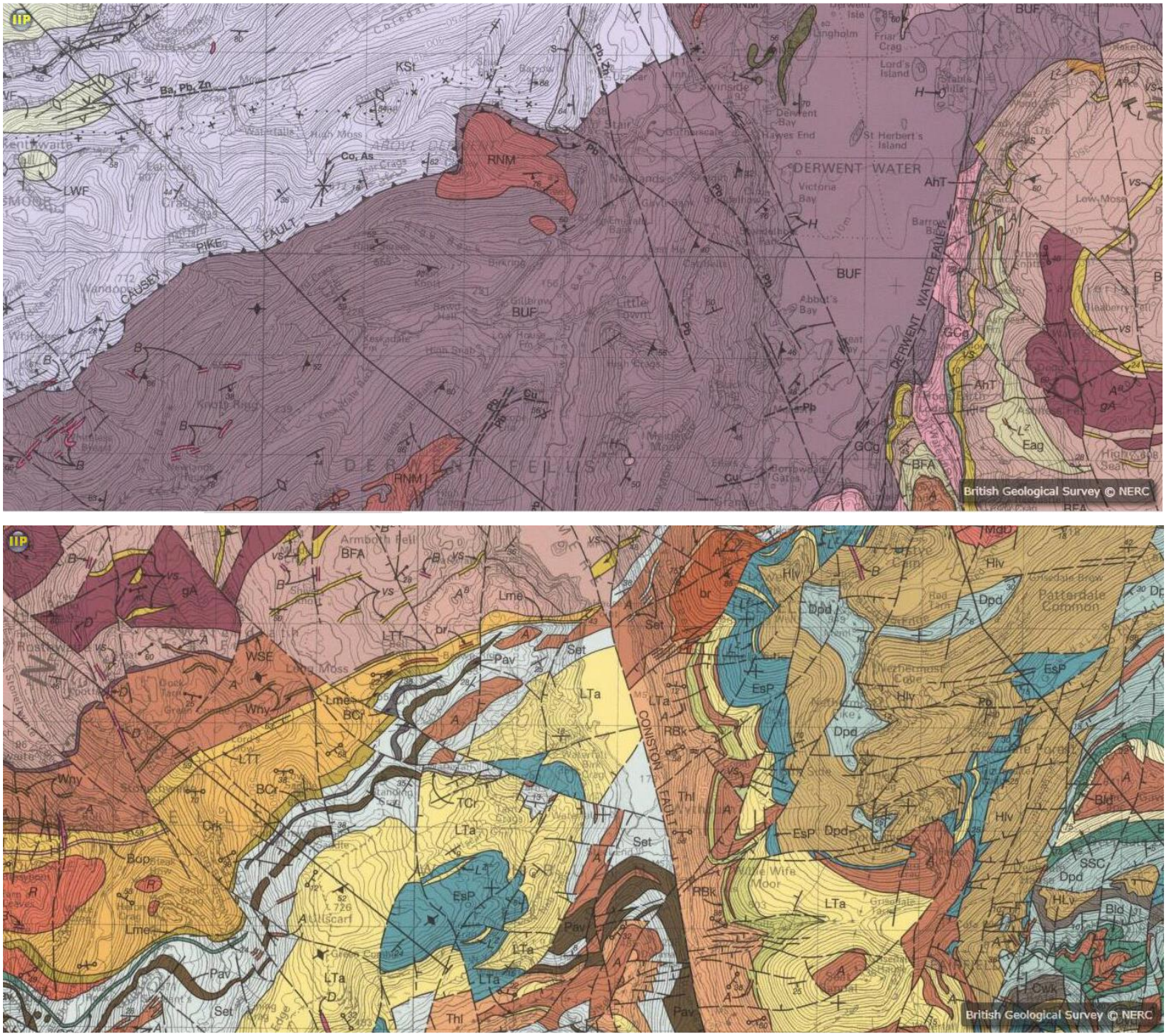


Figure 14. Cropped areas of interest from the BGS Keswick geological map.

Bonus Question to Test your Knowledge

If you have got this far, take a look at Figure X, below. This shows a sketched version of part of the Pembroke geological map. Using the dip directions to guide you, and the ages of the rocks you are given in the stratigraphic log, try to identify large-scale folded structures and highlight their location by drawing their axial trace.

You should also look at the numerous ~north–south trending faults displayed on the map, with thick black lines. Can you assess the direction and magnitude of displacement on these faults? Hint: the main displacement is horizontal not vertical. These are mostly right lateral, with offset around ~100-300 m, though there are some left lateral to be found; potentially a dip-slip component too.

